

TITLE: 50 GHz Vector Network Analyzer

Requesting Laboratory / Division / Group:

Communications Technology Laboratory
Radio Frequency Technology Division
Electromagnetic Fields Group (672.02)

General Statement of Need:

The existing Rohde and Schwarz model ZVA50 vector network analyzer currently in use and integrated into the NIST antenna laboratory calibration services and research has reached the end of service life and must be replaced. The replacement vector network analyzer must seamlessly fulfill the same technical requirements as and integrate with the existing equipment, meeting or exceeding all ZNA50 specifications. Demo models will be considered if manufacturer warranty and calibration services are included.

Item Specifications and Quantities:

Item 1:	Vector Network Analyzer	
Quantity:	1 EACH	
Specifications:	1.1	Direct access to the IF via front panel.
	1.2	Operate from 10 MHz up to 50 GHz.
	1.3	Direct access to the generator and receiver via front panel jumpers.
	1.4	4-test ports with 2.4 mm, male ruggedized connectors.
	1.5	Dedicated rear panel local oscillator (LO) access for external mmWave frequency converter LO.
	1.6	On-board Time Domain (TDR) analysis capability.
	1.7	Integrated capability to support the use of external Rohde and Schwarz mmWave frequency converters.
	1.8	Precision frequency reference with an aging per year of +/- 10 ⁻⁷ and temperature drift (between +5 degrees Celsius and +40 degrees Celsius) of +/-10 ⁻⁸ .
	1.9	2-way Wilkinson power divider that operates in the frequency range of 4GHz to 27GHz.

Vendor Requirements

1. Possess at time of award/contracting an Export Control Classification Number (ECCN).
2. Possess at time of award/contracting a Unique Entity Identification Number (UEI) which registration is listed as “Active” on the GSA System for Award Management (SAM.gov).

Procurement Support Information and Ancillary Services:

Section	Requirements/Specifications
Planning Considerations ☒ N/A	☐ List any visits required to allow potential vendors to see site specific constraints. [Explain Requirements.] ☐ US citizen required on-site (foreign nationals require pre-registration 30 days prior to visit.) ☐ Vendor Representative On-Site more than three (3) days. ☐ Other Considerations: [List other considerations to ensure an accurate quote.]
Shipping & Delivery ☐ N/A	Shipping Address: ☒ NIST-Boulder ATTN: TBA Mail Stop: TBA 325 Broadway Boulder, CO 80305 ☐ NIST-Gaithersburg ATTN: Mail Stop: 100 Bureau Dr Gaithersburg MD 20899 ☐ [Alternate Shipping Address] Shipping Criteria: ☐ Partial Delivery Acceptable ☐ [Enter Shipping and Delivery information by item.] ☒ Direct Delivery to OU Building Required This will be too big to get from the mail room. Building /Room Number: TBA ☐ Building has Loading Dock Containerization Preference: [Enter Container Type] Other Requirements: [List Any Other Requirement] Delivery Date Criteria:

☐ Delivery Shall Be Completed No Later Than:
Click or tap to enter a date.

☐ Delivery Shall Be Made No Earlier Than:
Click or tap to enter a date.

☐ Scheduled Deliveries:

Item Number	Due Date

☐ Other:
[List Other Shipping, Delivery or Special Requirements.]

Electronic Media

☒ N/A

☐ Software

☐ Electronic Manuals

☐ License Key

☐ Other:
[List Any Other Electronic Media Requirements]

☐ US Government Email Address for Software Delivery:
[Enter Address]

Installation

☒ N/A

☐ Vendor Technician on Site

☐ Rigging

☐ Uncrating / Unpackaging

☐ Removal of Packaging Material

☐ Equipment Set Up

☐ Start-Up Services

☐ Turnkey Installation

☐ Other:
[Enter Details]

NOTE: TPOC shall be the coordination POC.

Facility / Utility Considerations

☒ N/A

☐ [List Site Specific Utility Considerations by Item]

☐ OFPM Work Order Numbers
[List Work Order Numbers]

☐ Power Available

Voltage:

[Enter voltage] (V)

Maximum Current:

[Enter max current draw] (A)

Phase:

[Choose Phase]

Frequency:

[Typically 60Hz] (Hz)

NEMA Plug Type:

[Enter Plug Type] (i.e.: NEMA L5-20P)

Environment:

[Choose Environment]

Other Power Considerations:

[List Any Other Power Considerations.]

☐ Utility Connections:
[List Relevant Available Utilities, Pressure(s) and Flowrate(s).]

☐ Other Considerations:
[List Other Considerations to Ensure an Accurate Quote.]

Inspection and Acceptance

☐ N/A

☒ On-Site

☐ Acceptance Not Expected to Exceed 7 Business Days

☒ Acceptance Expected to Exceed 7 Business Days:
Three Weeks After Delivery of Order

☒ Inspection and Acceptance Plan:
A set of validation measurements will be performed.

☐ Other:
[Enter Other Requirements]

Training

☒ N/A

Training Requirements (Who, When, What, Where)

☐ Number of Trainees:

[Enter Number of Trainees]

☐ On-Site

☐ Off-Site

Location: [Enter Location(s)]

Is Trainee Travel Required:

[Choose Answer]

Scope of Training

☐ Operation

☐ Maintenance

☐ Troubleshooting

☐ Safety

☐ Other

[Enter Other Scope of Training Requirements]

☐ Training Materials to be Provided by Government:
[Describe Materials]

☐ Training Materials to be Provided by Vendor:
[Describe Materials]

☐ Other
[Enter Other Training Requirements]

NOTE: SME shall be coordination POC.	
Warranty ☐ N/A	Extended Warranty Required Warranty Length: 2-Years What is included? ☒ Labor ☒ Parts ☐ Travel ☐ Cost and Liability for Returns ☒ Other 2-Year Extended Manufacturer Warranty Vendor Option WE2ZNA50 WARR EXT 2Y ☐ 1 Option Period ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Calibration ☐ N/A	[Describe Calibration Requirements Per Item] Calibration Plan Length: 3- Years What is Included? ☐ Labor ☐ Spare Parts ☐ Consumables ☐ Travel ☐ Cost and Liability for Shipment ☐ Software Updates ☐ Manuals ☒ Other 3-Year Extended Calibration Plan R&S Option CC3ZNA50 Cal Cov 3Y ☒ Requirements: Two scheduled calibrations. ☐ 1 Option Period Length: [i.e. 1 – Year / 6 - Months] ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Maintenance ☒ N/A	[Describe Maintenance Requirements Per Item] Maintenance Plan Length: [i.e. 3 - Years] What is Included? ☐ Labor ☐ Spare Parts ☐ Consumables ☐ Travel ☐ Cost and Liability for Shipment ☐ Software Updates ☐ Manuals ☐ Other [Enter Other Requirements] ☐ Requirements: [Enter Requirements] (i.e.: On-Site Within 72 hours, etc.) ☐ 1 Option Period Length: [i.e. 1 – Year / 6 - Months] ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Government Furnished Material / Equipment ☒ N/A	List Any Government Furnished Material / Equipment Per Item] ☐ Samples for Testing ☐ Equipment which will leave Government Site Description: [Enter Description] Serial Number: [Enter Serial Number] NIST Property Number: [Enter NIST Property Number] ☐ Other: [Enter Other Details]
Travel ☒ N/A	[Describe any Travel Requirements Per Item] ☐ Explain: [Enter Justification / Requirement] ☐ Other: [Enter Other Requirements] NOTE TO VENDOR: To the maximum extent possible, include travel requirement estimates as part of pertinent sections above (Installation, Maintenance, Repair, etc.). Otherwise, Travel will be a separate line item, reimbursable per the Federal Travel Regulation.